Predicting Housing Price Variables for California, 1990

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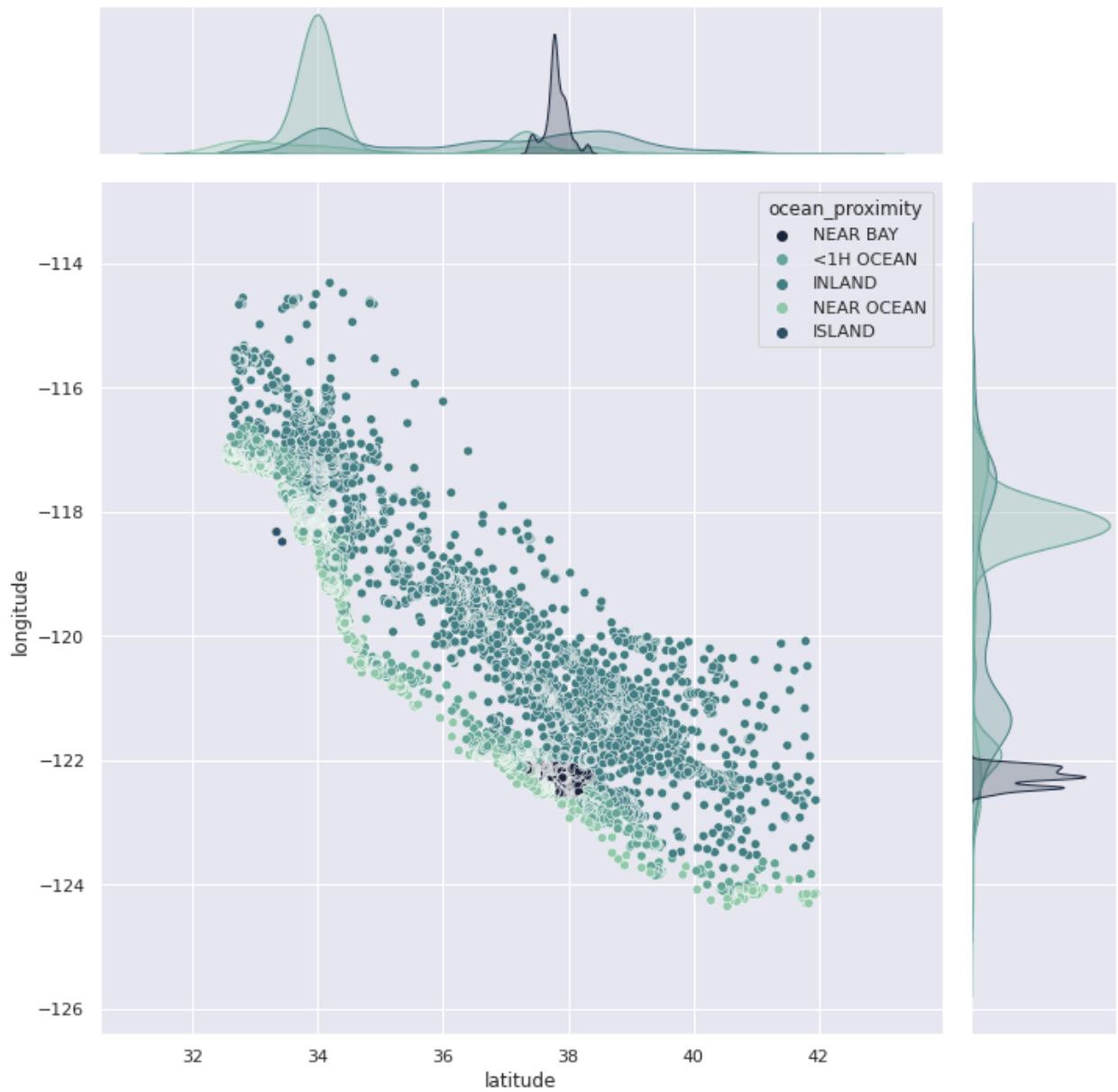
I. Introduction

The California Housing dataset is a collection of median house prices and other attributes that could influence the price collected from the 1990 census. There are a total of 10 different columns. Just to get started with a rough idea of what the dataset entails, I plotted a map of latitude vs. longitude using seaborn and split it by color into its proximity to the ocean.

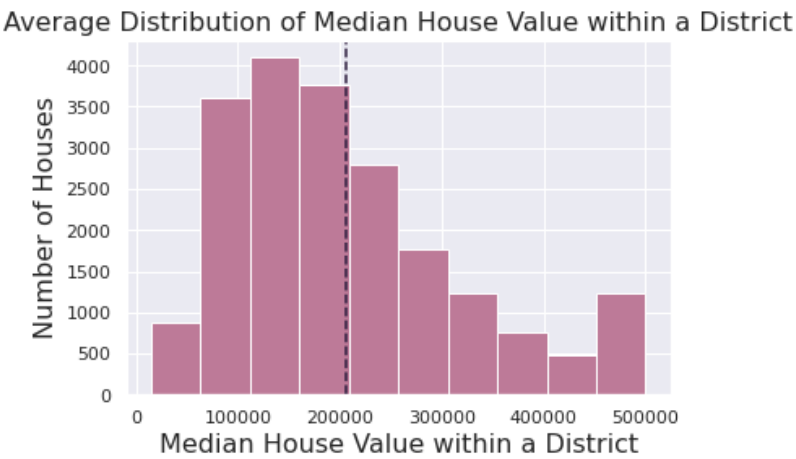


Although it creates a nice visualization of the collection of houses gathered for this dataset, we cannot gather enough information on how this affects housing prices yet.

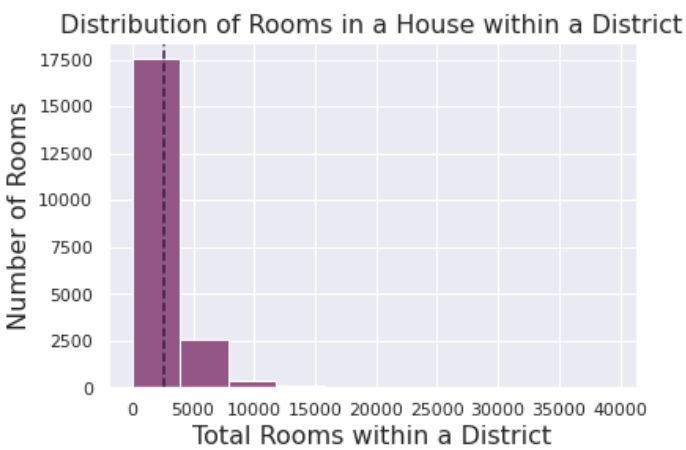
II. Data Cleaning and Variable Selection:

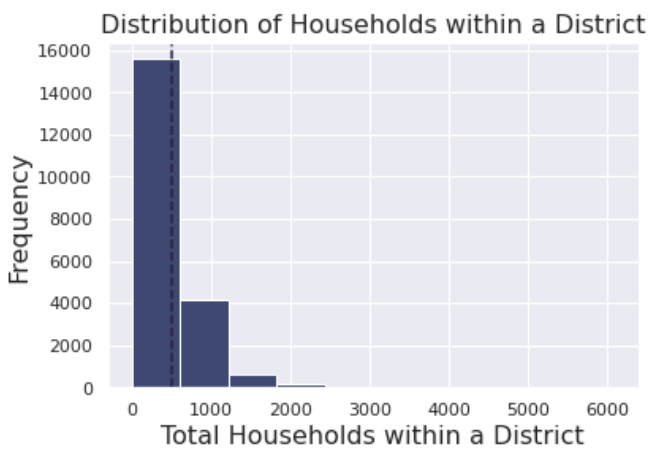
Let's continue to actually define the important attributes in this next step and see whether the above plot matters in determining housing prices. The data cleaning process was fairly simple as there weren't any missing values, duplicates, nor unknown values. The attribute names were also small and understandable enough to use, and no changes or transformations were deemed necessary. Since we are building a model to predict housing prices, we can set the target variable as "housing_median_value", and start to narrow down the other attributes by checking normal distributions and correlations to the target variable. Before even plotting, we can infer what attributes are more likely to have a direct correlation with the housing price. Out of the other 9 dependent variables, we can make a guess that the "median_income" is probably going to have the highest rate of influence directly with the price of a house.

Starting with the normal distributions, I initially used histograms, along with calculating the mean to check whether the variables were normally distributed or not.



After plotting a histogram and labeling the mean on the target variable, we can say it's relatively normally distributed. So we can move onto the attributes.

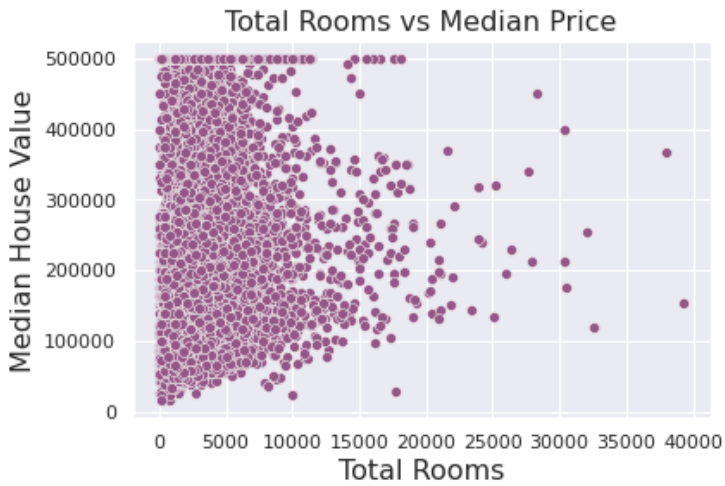


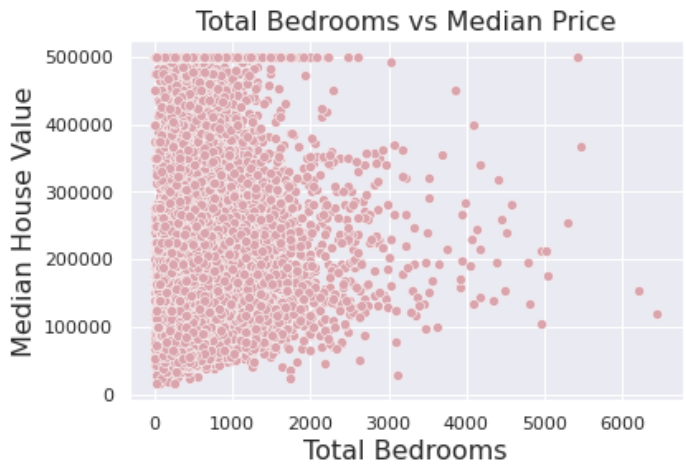


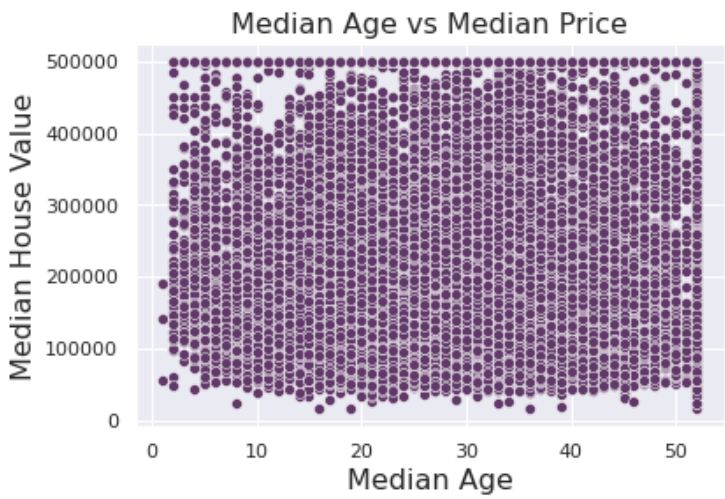
I haven't included the distributions for every column in this report, for the sake of space, but they're all available in my code. Several distributions of the attributes are skewed to the right because of outliers, but most of the numbers are still heavily concentrated within one region. The

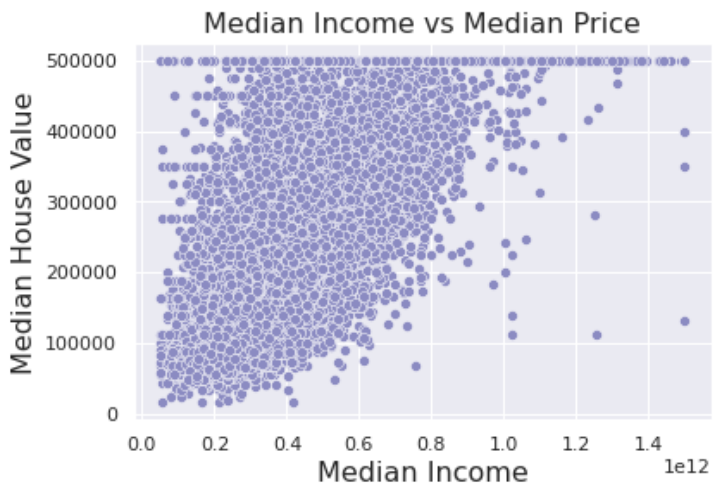
idea of standardizing/normalizing these attributes seems unnecessary because of the nature of the variables. So, I have decided to leave it as it is.

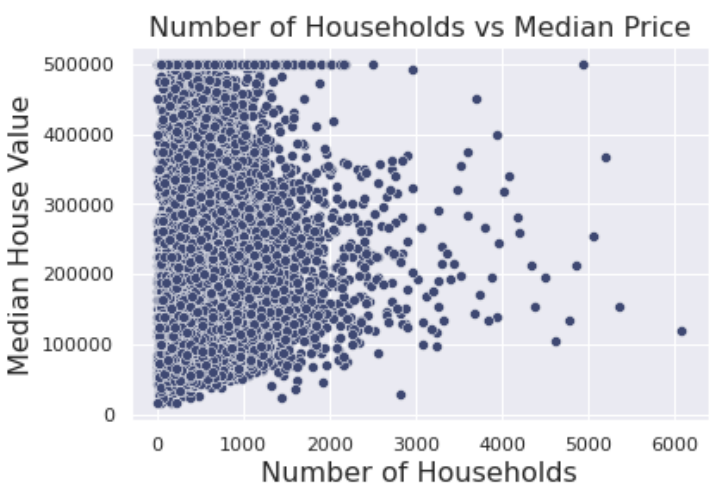
Furthermore, I used scatterplots to figure which variables had the most influence on the median value of households.



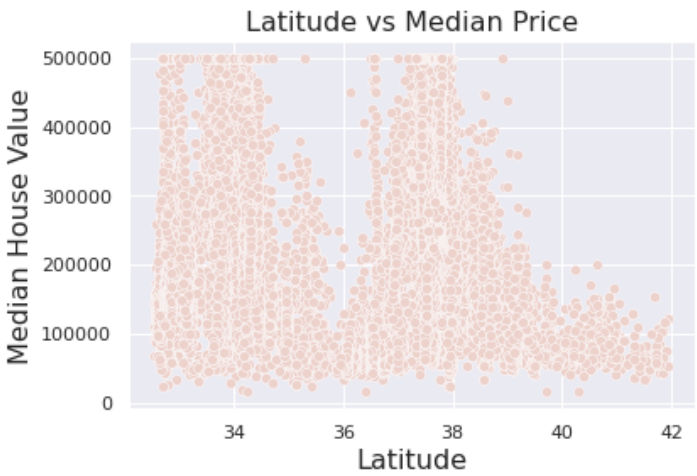


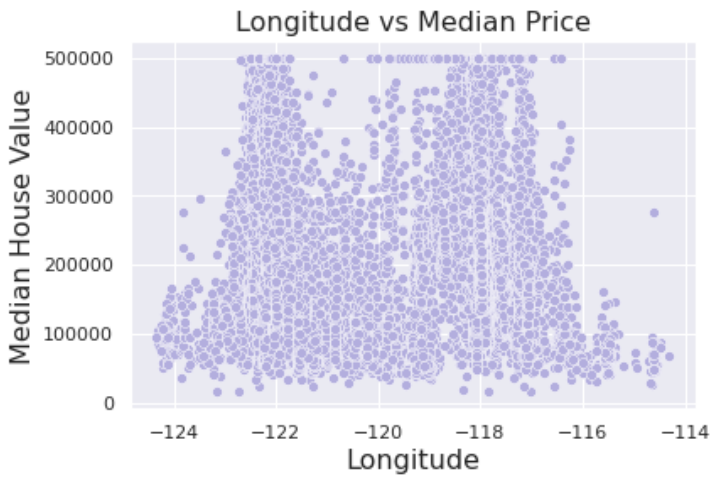


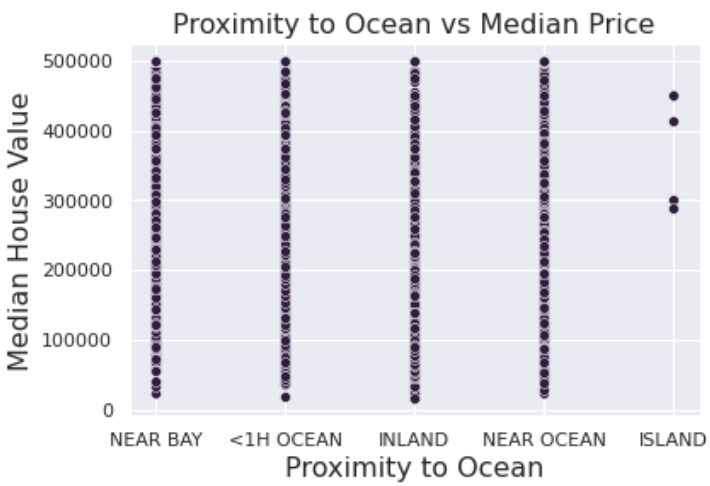




The scatterplots yield some interesting results. As I predicted before, the median income seems to have the most direct positive relationship- as income increases, the median house price also increases. The other variables that show somewhat of a correlation are the rooms, bedrooms, households and population. The plot of median age vs median price is something that might not be expected. However, we can infer that although the price of land appreciates over time, if a house has not gone under much maintenance, it would still sell for a low price. Thus, the points are all over the plot and there is no clear correlation.







Now, looking at the longitude and latitude, an interesting observation can be made. The highest price range seems to occur around (34, -118) and (38, -122). Looking back at the initial geographical map I plotted, we can see that the cluster of points that would be considered near the bay occurs at (38, -122). So, although the scatter plot of ocean proximity doesn't show any clear correlation with housing value, putting these 3 attributes together, we can make the inference that some of the highest housing prices from the state of California come from near the bay.

To keep it simple, I have decided to further proceed with only median price, total rooms, and number of households within a district as the chosen attributes.

III. Creating a Linear Regression model

Using the variables I selected in the previous step, I split the data into test and train sets with 0.2 and 0.8 sizes respectively. Running the LR model to find some of the statistics, I found that in the training set, the RMSE(Root Mean Square Error) is 81267.9018 and the R2(R-Squared Test) score is 0.4994. For the testing set the RMSE is 82574.0006 and the R2 score is 0.5058. In addition I ran an Ordinary Least Squares Test to find the overall summary between correlation status.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:     median_house_value   R-squared:                       0.499
Model:                            OLS   Adj. R-squared:                  0.499
Method:                 Least Squares   F-statistic:                     5490.
Date:                Wed, 22 Sep 2021   Prob (F-statistic):               0.00
Time:                        04:23:47   Log-Likelihood:            -2.1011e+05
No. Observations:               16512   AIC:                         4.202e+05
Df Residuals:                   16508   BIC:                         4.203e+05
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const       2.065e+05    632.520    326.465      0.000    2.05e+05    2.08e+05
0           8.895e+04    721.288    123.327      0.000    8.75e+04    9.04e+04
1          -5.191e+04   1794.258    -28.931      0.000   -5.54e+04   -4.84e+04
2           5.474e+04   1768.850     30.948      0.000    5.13e+04    5.82e+04
==============================================================================
Omnibus:                     3456.565   Durbin-Watson:                   1.997
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             8340.262
Skew:                           1.167   Prob(JB):                         0.00
Kurtosis:                       5.584   Cond. No.                         5.52
==============================================================================
```

The OLS test shows that all the variables selected have a low P-value which indicates high correlation, with the first variable(median income) having the lowest rate of standard error. However, since the accuracy still has room for improvement I decided to include latitude and longitude and rerun the linear regression model. Now, the training RMSE is 73799.9865 and R2 score is 0.5872. The testing RMSE is 74696.6171 and R2 score is 0.5956. The accuracy of the model has clearly improved after adding those variables.

IV. Conclusion

Overall, it looks like the more significant variables you add to the regression model, the easier it will be to build a linear regression model that can predict the price based on a given pattern. The initial observation based on the scatterplots was mostly accurate. The median income will obviously have the highest influence on determining the price of the house that will be bought. Also, after adding latitude and longitude, we can say that the houses close to the bay area are generally the most expensive in California.

Code

Github Code Link:
https://github.com/abhii99/basic-projects/blob/main/California-Housing/1990_California_Housing_Code.ipynb

Python Code:

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
from sklearn.metrics import r2_score
from sklearn.datasets import load_boston
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
from sklearn.linear_model import SGDRegressor
from sklearn.preprocessing import StandardScaler

df =
pd.read_csv("https://utdallas.box.com/shared/static/nhj8rs6qmyw9k16xkcjzgzgkc6t5hpla.csv")

df = df.drop_duplicates()
df.duplicated().values.any() # Checking for an duplicate entries

df = df.fillna(method="ffill")
pd.isnull(df).any()  # Checking for Null Values

df.head()

#plotting lattitude vs lattitude
plt.figure(figsize=(10,10))
sns.jointplot(x=df.latitude, y=df.longitude, hue=df.ocean_proximity,
            palette=('#1c273e', '#64a698', '#457f85','#91c9ac', '#2f5269'),size = 10)
plt.show()
sns.despine

sns.set()
plt.hist(df.median_house_value, color='#bd7a98')
```

```
plt.xlabel('Median House Value within a District', fontsize=16)
plt.ylabel('Number of Houses', fontsize=16)
plt.title('Average Distribution of Median House Value within a District', fontsize=16)

plt.axvline(df.median_house_value.mean(), color='#2d1e3e', linestyle='dashed')

mean_house_value = df["median_house_value"].mean()
print(mean_house_value)

plt.hist(df.total_rooms, color='#945785')
plt.xlabel('Total Rooms within a District', fontsize=16)
plt.ylabel('Number of Rooms', fontsize=16)
plt.title('Distribution of Rooms in a House within a District', fontsize=16)

plt.axvline(df.total_rooms.mean(), color='#2d1e3e', linestyle='dashed')

ax = sns.scatterplot(x="total_rooms", y="median_house_value", data=df,color='#945785')
ax.set_title('Total Rooms vs Median Price', fontsize=16)
ax.set_xlabel("Total Rooms", fontsize=16)
ax.set_ylabel("Median House Value", fontsize=16)

plt.hist(df.total_bedrooms, color='#daa4ac')
plt.xlabel('Total Rooms within a District', fontsize=16)
plt.ylabel('Number of Rooms', fontsize=16)
plt.title('Average Distribution of Median House Value within a District', fontsize=16)

plt.axvline(df.total_bedrooms.mean(), color='#2d1e3e', linestyle='dashed')

ax = sns.scatterplot(x="total_bedrooms", y="median_house_value", data=df, color='#daa4ac')
ax.set_title('Total Bedrooms vs Median Price', fontsize=16)
ax.set_xlabel("Total Bedrooms", fontsize=16)
ax.set_ylabel("Median House Value", fontsize=16)

plt.hist(df.housing_median_age, color='#613969')
plt.xlabel('Total Rooms within a District', fontsize=16)
plt.ylabel('Number of Rooms', fontsize=16)
plt.title('Average Distribution of Median House Value within a District', fontsize=16)

plt.axvline(df.housing_median_age.mean(), color='#2d1e3e', linestyle='dashed')
```

```
ax = sns.scatterplot(x="housing_median_age", y="median_house_value", data=df,
color='#613969')
ax.set_title('Median Age vs Median Price', fontsize=16)
ax.set_xlabel("Median Age", fontsize=16)
ax.set_ylabel("Median House Value", fontsize=16)

df.median_income = df.median_income
plt.hist(df.median_income, color='#8b8bc2')
plt.xlabel('Total Rooms within a District', fontsize=16)
plt.ylabel('Number of Rooms', fontsize=16)
plt.title('Average Distribution of Median House Value within a District', fontsize=16)

plt.axvline(df.median_income.mean(), color='#2d1e3e', linestyle='dashed')

ax = sns.scatterplot(x="median_income", y="median_house_value", data=df, color='#8b8bc2')
ax.set_title('Median Income vs Median Price', fontsize=16)
ax.set_xlabel("Median Income", fontsize=16)
ax.set_ylabel("Median House Value", fontsize=16)

plt.hist(df.households, color='#3e4872')
plt.xlabel('Total Households within a District', fontsize=16)
plt.ylabel('Frequency', fontsize=16)
plt.title('Distribution of Households within a District', fontsize=16)

plt.axvline(df.households.mean(), color='#2d1e3e', linestyle='dashed')

ax = sns.scatterplot(x="households", y="median_house_value", data=df, color='#3e4872')
ax.set_title("Number of Households vs Median Price", fontsize=16)
ax.set_xlabel('Number of Households', fontsize=16)
ax.set_ylabel("Median House Value", fontsize = 16)

plt.hist(df.population, color='#646a9f')
plt.xlabel('Total Rooms within a District', fontsize=16)
plt.ylabel('Number of Rooms', fontsize=16)
plt.title('Average Distribution of Median House Value within a District', fontsize=16)

plt.axvline(df.population.mean(), color='#2d1e3e', linestyle='dashed')

ax = sns.scatterplot(x="population", y="median_house_value", data=df, color='#646a9f')
ax.set_title('Population vs Median Price', fontsize=16)
```

```
ax.set_xlabel("Population", fontsize=16)
ax.set_ylabel("Median House Value", fontsize=16)

plt.hist(df.ocean_proximity, color='#2d1e3e')
plt.xlabel('Proximity to Ocean', fontsize=16)
plt.ylabel('Frequency', fontsize=16)
plt.title('Distribution of Houses Based on Proximity to Ocean', fontsize=16)

ax = sns.scatterplot(x="ocean_proximity", y="median_house_value", data=df,color='#2d1e3e')
ax.set_title('Proximity to Ocean vs Median Price', fontsize=16)
ax.set_xlabel("Proximity to Ocean", fontsize=16)
ax.set_ylabel("Median House Value", fontsize=16)

ax = sns.scatterplot(x="longitude", y="median_house_value", data=df, color='#b3aede')
ax.set_title('Longitude vs Median Price', fontsize=16)
ax.set_xlabel("Longitude", fontsize=16)
ax.set_ylabel("Median House Value", fontsize=16)

ax = sns.scatterplot(x="latitude", y="median_house_value", data=df,color='#edd1cb')
ax.set_title('Latitude vs Median Price', fontsize=16)
ax.set_xlabel("Latitude", fontsize=16)
ax.set_ylabel("Median House Value", fontsize=16)

#target variable: median_house_value
#some attributes to consider: median_income, households, total_rooms, total_bedrooms

#setting the dependent and independent variables based on my selection
X = df[["median_income", "total_rooms", "households"]]
Y = df['median_house_value']

s = StandardScaler()
X = pd.DataFrame(s.fit(X).fit_transform(X))

#splitting into test and train groups
from sklearn.model_selection import train_test_split

X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size = 0.2, random_state=5)
print(X_train.shape)
print(X_test.shape)
print(Y_train.shape)
```

```
print(Y_test.shape)

from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

lin_model = LinearRegression()
lin_model.fit(X_train, Y_train)

y_train_predict = lin_model.predict(X_train)
rmse = (np.sqrt(mean_squared_error(Y_train, y_train_predict)))
r2 = r2_score(Y_train, y_train_predict)

print("The model performance for training set")
print("--------------------------------------")
print('RMSE is {}'.format(rmse))
print('R2 score is {}'.format(r2))

y_test_predict = lin_model.predict(X_test)
rmse = (np.sqrt(mean_squared_error(Y_test, y_test_predict)))
r2 = r2_score(Y_test, y_test_predict)

print("The model performance for testing set")
print("--------------------------------------")
print('RMSE is {}'.format(rmse))
print('R2 score is {}'.format(r2))

import statsmodels.api as sm
from statsmodels.sandbox.regression.predstd import wls_prediction_std

X_train_off = sm.add_constant(X_train)
smRegression = sm.OLS(Y_train, X_train_off)

model = smRegression.fit()
print(model.summary())

X = df[["median_income", "total_rooms", "households", "latitude", "longitude"]]
Y = df['median_house_value']

s = StandardScaler()
X = pd.DataFrame(s.fit(X).fit_transform(X))
```

```
#splitting into test and train groups
from sklearn.model_selection import train_test_split

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smRegression = sm.OLS(Y_train, X_train_off)

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```